TauTimp — Entropy Virtual Particle: Variable-Coefficient Wave Equation
Onduscular Theory (Teoria Onduscular) — Viorel Popescu, 25April 2026
(v3 — corrected Maple errors: frac → frac_v; surfdata → matrixplot; zgradient syntax fixed)

1. Master Equation and Physical Setup

The entropy virtual-particle field $S(r,\theta,t)$ satisfies the TauTimp onduscular wave equation:

$$\frac{R}{r}\,\Delta S(r,\theta,t) = \frac{1}{V^2}\frac{\partial^2 S}{\partial t^2}$$

The propagation velocity V depends on the virtual-particle species (transmitted as a **vacancy/hole**, not a classical particle):

Virtual Particle	Velocity V	Physical analogy
Photon	$c = 2.998 \times 10^8$ m/s	electromagnetic void
Electron	$v_e = c/137$ (Bohr orbital)	electronic vacancy
Phonon (acoustic)	$v_s = 346$ m/s	acoustic vacancy
Gamma / transverse	$V_\gamma = 10^9\, c$	transverse field quantum

Key parameter: $R = 4.2355 \times 10^{22}$ m (TauTimp cosmological radius).

We solve the prototype with $V = c$:

$$\boxed{\frac{R}{r}\,\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}}$$

2. Spherical Symmetry — Radial Laplacian

For $f = f(r,t)$ (no θ,ϕ dependence, $\ell = 0$ sector):

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)$$

The PDE becomes:

$$\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}\right) = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}$$

3. Separation of Variables: $f(r,t) = g(r)\,u(t)$

Substituting and dividing by $g(r)\,u(t)$:

$$\frac{R}{r}\cdot\frac{g'' + \frac{2}{r}g'}{g} = \frac{\ddot{u}}{c^2\,u} = -\omega^2$$

3.1 Temporal ODE

$$\ddot{u} + \omega^2 c^2\,u = 0 \Longrightarrow \boxed{u(t) = A\cos(\omega c\,t) + B\sin(\omega c\,t)}$$

3.2 Radial ODE

$$g'' + \frac{2}{r}g' + \frac{\omega^2 r}{R}\,g = 0$$

Reduction: Set $g(r) = \frac{\psi(r)}{r}$. Then:

$$g' = \frac{\psi' r - \psi}{r^2}, \qquad g'' = \frac{\psi'' r^2 - 2\psi' r + 2\psi}{r^3}$$

Substituting into (ROD):

$$\frac{\psi''}{r} - \frac{2\psi'}{r^2} + \frac{2\psi}{r^3} + \frac{2}{r}\left(\frac{\psi'}{r} - \frac{\psi}{r^2}\right) + \frac{\omega^2}{R}\psi = 0$$

All terms except ψ''/r and $(\omega^2/R)\psi$ cancel exactly:

$$\frac{\psi''}{r} + \frac{\omega^2}{R}\psi = 0 \Longrightarrow \boxed{\psi'' + \frac{\omega^2}{R}\,\psi = 0}$$

This is **simple harmonic oscillator** in r. With $k = \omega/\sqrt{R}$:

$$\psi(r) = C_1\cos(kr) + C_2\sin(kr)$$
$$\Longrightarrow g(r) = \frac{C_1\cos(kr) + C_2\sin(kr)}{r}$$

Regularity at $r \to 0$: The cosine term gives $g \sim C_1/r \to \infty$, so $C_1 = 0$:

$$\boxed{g(r) = \frac{C_2\,\sin(kr)}{r}, \qquad k = \frac{\omega}{\sqrt{R}}}$$

4. Boundary Condition and Eigenvalue Spectrum

Condition: $f(2R,\,0) = 0$ with $u(0) = 1 \Rightarrow g(2R) = 0$:

$$\frac{C_2\,\sin(2k_n R)}{2R} = 0 \Longrightarrow \sin(2k_n R) = 0 \Longrightarrow 2k_n R = n\pi, \quad n = 1,2,3,\ldots$$

$$\boxed{k_n = \frac{n\pi}{2R}, \qquad \omega_n = k_n\sqrt{R} = \frac{n\pi}{2\sqrt{R}}}$$

Numerical values ($R = 4.2355 \times 10^{22}$ m, $\sqrt{R} = 6.5080 \times 10^{11}$ m$^{1/2}$):

n	k_n (m^{-1})	ω_n (rad/s)	$T_n = 2\pi/(\omega_n c)$ (s)
1	3.709×10^{-23}	2.414×10^{-12}	8.683×10^{3}
2	7.417×10^{-23}	4.827×10^{-12}	4.342×10^{3}
3	1.113×10^{-22}	7.241×10^{-12}	2.894×10^{3}

5. Complete Solution $f(r,t)$

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} \frac{\sin(k_n r)}{r}\left[A_n \cos(\omega_n c\, t) + B_n \sin(\omega_n c\, t)\right]}$$

Fundamental mode snapshot:

$$f_1(r,0) = A_1\,\frac{\sin(k_1 r)}{r}$$

Fundamental oscillation period (photon, $V = c$):

$$T_1 = \frac{2\pi}{\omega_1 c} = \frac{4\sqrt{R}}{c} \approx 8683\ s \approx 2.41 \text{ hours}$$

6. Angular Extension: Building $S(r,\theta,t)$

For the full equation with azimuthal symmetry ($m = 0$), the angular part satisfies the Legendre equation with eigenvalue $\ell(\ell+1)$. The complete entropy field:

$$S(r,\theta,t) = \sum_{n=1}^{\infty}\sum_{\ell=0}^{\infty} \frac{\sin(k_n r)}{r}\, P_\ell(\cos\theta)\left[A_{n\ell}\cos(\Omega_{n\ell} t) + B_{n\ell}\sin(\Omega_{n\ell} t)\right]$$

For the spherically symmetric sector ($\ell = 0$, $P_0 = 1$): $S(r,\theta,t) = f(r,t)$.

7. Probabilistic Entropy Flux — Field $P(r,t)$ 7.1 d'Alembert Equation

The entropy phonon probability density satisfies:

$$\frac{1}{c^2}\frac{\partial^2 P}{\partial t^2} - \frac{\partial^2 P}{\partial r^2} = 0$$

with flux density $\mathcal{F}(r,t) = P(r,t)/(4\pi r^2)$.

7.2 Boundary and Initial Conditions

- $P(2R,\ 0) = 0$ (zero flux at the onduscular boundary)
- $P(r,t)$ is **degree-2 polynomial in** r modulated by a **Gaussian** (at $t = 0$)
- $\frac{\partial P}{\partial r}|_{r=0,\,t=0} = R$ (finite, nonzero gradient at origin — the "hole" injection rate)

7.3 Ansatz at $t = 0$

We seek:

$$P(r,0) = (a + b\,r + c_2\,r^2)\exp\left(-\frac{r^2}{2\sigma^2}\right)$$

Apply $P(2R,0) = 0$**:**

$$(a + 2bR + 4c_2R^2)\,e^{-2R^2/\sigma^2} = 0 \Longrightarrow a + 2bR + 4c_2R^2 = 0$$

Apply $\partial_r P|_{r=0} = R$**:**

$$\frac{\partial P}{\partial r}\Big|_{r=0} = b\,e^{0} = b = R \tag{BC2}$$

From (BC2): $b = R$. Substituting into (BC1): $a = -2R^2 - 4c_2R^2$.

Choose $c_2 = -1$ (quadratic decay, normalization absorbed into $\mathcal{N}$):

$$a = -2R^2 + 4R^2 = 2R^2$$

Therefore the initial profile is:

$$\boxed{P(r,0) = \mathcal{N}(2R^2 + Rr - r^2)\exp\left(-\frac{r^2}{2R^2}\right)}$$

Verification of conditions:

$$P(2R,0) = \mathcal{N}(2R^2 + 2R^2 - 4R^2)\,e^{-2} = \mathcal{N}\cdot 0\cdot e^{-2} = 0\checkmark$$

$$\frac{\partial P}{\partial r}\Big|_{r=0} = \mathcal{N}[R - 0]\cdot 1 = \mathcal{N}\,R \overset{!}{=} R \Longrightarrow \mathcal{N} = 1$$

So the normalization constant is fixed exactly: $\mathcal{N} = 1$, and:

$$\boxed{P(r,0) = (2R^2 + Rr - r^2)\exp\left(-\frac{r^2}{2R^2}\right)}$$

7.4 Factored Form

The polynomial $2R^2 + Rr - r^2 = -(r^2 - Rr - 2R^2) = -(r - 2R)(r + R)$, so:

$$\boxed{P(r,0) = -(r-2R)(r+R)\exp\left(-\frac{r^2}{2R^2}\right)}$$

For $0 \le r \le 2R$: $(r-2R) \le 0$ and $(r+R) > 0$, so $P(r,0) \ge 0$ throughout the domain. ✓

7.5 d'Alembert General Solution

From equation (dA) with $\partial_t P(r,0) = 0$ (standing-wave initial condition):

$$P(r,t) = \frac{1}{2}[h(r-ct) + h(r+ct)]$$

where $h(\xi) = -(\xi-2R)(\xi+R)\exp\left(-\frac{\xi^2}{2R^2}\right)$.

Explicit full solution:

$$\boxed{P(r,t) = \frac{1}{2}\left[-(r-ct-2R)(r-ct+R)\,e^{-(r-ct)^2/(2R^2)} - (r+ct-2R)(r+ct+R)\,e^{-(r+ct)^2/(2R^2)}\right]}$$

7.6 Flux Density

$$\mathcal{F}(r,t) = \frac{P(r,t)}{4\pi r^2} = \frac{-(r-ct-2R)(r-ct+R)\,e^{-(r-ct)^2/(2R^2)} + (\text{retarded})}{8\pi r^2}$$

At $t = 0$:

$$\mathcal{F}(r,0) = \frac{-(r-2R)(r+R)}{4\pi r^2}\,e^{-r^2/(2R^2)}$$

8. Normalization of $P(r,t)$

Require total probability unity:

$$\int_0^{2R} \frac{P(r,0)}{4\pi r^2}\cdot 4\pi r^2\,dr = \int_0^{2R} P\,(r,0)\,dr = 1$$

$$\int_0^{2R} -(r-2R)(r+R)\,e^{-r^2/(2R^2)}\,dr = \int_0^{2R} (\,2R^2 + Rr - r^2)\,e^{-r^2/(2R^2)}\,dr$$

Splitting ($u = r/(\sqrt{2}R)$, $dr = \sqrt{2}R\,du$, upper limit $u_{\max} = \sqrt{2}$):

$$I_0 = 2R^2\int_0^{2R} e^{-r^2/(2R^2)}\,dr = 2R^2\cdot\sqrt{2}R\cdot\frac{\sqrt{\pi}}{2}\,\mathrm{erf}(\sqrt{2}) = \sqrt{2\pi}\,R^3\,\mathrm{erf}(\sqrt{2})$$

$$I_1 = R\int_0^{2R} r\;e^{-r^2/(2R^2)}\,dr = R\cdot R^2(1-e^{-2}) = R^3(1-e^{-2})$$

$$I_2 = \int_0^{2R} r^2\;e^{-r^2/(2R^2)}\,dr = R^3\left[\sqrt{\frac{\pi}{2}}\,\mathrm{erf}(\sqrt{2}) - \sqrt{2}\,e^{-2}\right]$$

Total: $I = I_0 + I_1 - I_2$. Using $\mathrm{erf}(\sqrt{2}) \approx 0.9953$, $e^{-2} \approx 0.1353$:

$$I_0 \approx \sqrt{2\pi}\cdot R^3\cdot 0.9953 \approx 2.4944\,R^3$$

$$I_1 \approx R^3(1-0.1353) = 0.8647\,R^3$$

$$I_2 \approx R^3\left[\sqrt{\pi/2}\cdot 0.9953 - \sqrt{2}\cdot 0.1353\right] \approx R^3[0.8862\cdot 0.9953 - 1.4142\cdot 0.1353] \approx R^3[0.8820 - 0.1913]$$
$$= 0.6907\,R^3$$

$$\boxed{I = I_0 + I_1 - I_2 \approx (2.4944 + 0.8647 - 0.6907)\,R^3 \approx 2.6684\,R^3}$$

If the physical normalization requires $\int P\,dr = 1$, then $\mathcal{N} = 1/I \approx 0.3748/R^3$.

With $R = 4.2355\times 10^{22}$ m, $R^3 = 7.595\times 10^{67}$ m^3:

$$\mathcal{N}_{\text{phys}} \approx 4.93\times 10^{-70}\ \text{m}^{-3}$$

9. Peak Position and Width of Flux Profile

At $t = 0$, the flux $\mathcal{F}(r,0) \propto (2R^2 + Rr - r^2)\,e^{-r^2/(2R^2)}$.

Setting $d\mathcal{F}/dr = 0$:

$$(R-2r)\,e^{-r^2/(2R^2)} + (2R^2 + Rr - r^2)\left(-\frac{r}{R^2}\right)e^{-r^2/(2R^2)} = 0$$

$$(R-2r) - \frac{r(2R^2 + Rr - r^2)}{R^2} = 0$$

$$R^3 - 2R^2r - 2R^2r - Rr^2 + r^3 = 0 \Rightarrow r^3 - Rr^2 - 4R^2r + R^3 = 0$$

Roots (numerical, $\rho = r/R$): $\rho^3 - \rho^2 - 4\rho + 1 = 0$.

$\rho_1 \approx 0.246$ $(r \approx 0.246R)$, $\rho_2 \approx -1.802$ (unphysical), $\rho_3 \approx 2.556$ $(r > 2R,$ outside domain)

Peak of flux at $r^* \approx 0.246\,R = 1.04\times 10^{22}$ **m** (interior of the onduscular sphere).

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10. Summary Table

Quantity	Formula	Value
R	given	4.2355×10^{22} m

Quantity	Formula	Value
$\sqrt{R}$	$\sqrt{R}$	6.508×10^{11} m$^{1/2}$
k_1	$\pi/(2R)$	3.709×10^{-23} m^{-1}
ω_1	$\pi/(2\sqrt{R})$	2.414×10^{-12} rad/s
T_1 (photon)	$4\sqrt{R}/c$	$8683\ s \approx 2.41$ h
$P(r,0)$	$-(r-2R)(r+R)e^{-r^2/(2R^2)}$	Gaussian-quadratic
$\partial_r P\|_{r=0}$	R (BC)	4.2355×10^{22} m
$P(2R,0)$	0 (BC)	exact
Flux peak	$r^* \approx 0.246R$	1.04×10^{22} m
$\mathcal{N}_{\text{phys}}$	$1/I$	4.93×10^{-70} m^{-3}

11. Connection Between $f(r,t)$ and $P(r,t)$

The probability density field and the entropy wave are linked by the squared-amplitude relation standard in TauTimp:

$$P(r,t) \propto |f(r,t)|^2 \cdot 4\pi r^2$$

For the fundamental mode:

$$P_1(r,t) = \mathcal{N} \sin^2(k_1 r)[A_1 \cos(\omega_1 ct) + B_1 \sin(\omega_1 ct)]^2$$

This automatically satisfies $P(2R,\cdot) = 0$ since $\sin(k_1 \cdot 2R) = \sin(\pi) = 0$. The d'Alembert form captures the wave propagation of this probability density.

12. Maple Implementation Notes (v3 Bug Fixes)

Three errors were corrected from the v2 Maple transcript:

Error 1 — Protected name frac.

Maple protects the name frac (it is a built-in function). The loop variable was renamed to frac_v in every for frac_v in [...] loop.

Error 2 — surfdata axis-list syntax.

surfdata(Matrix, list1, list2, ...) does **not** accept two bare Maple lists as positional axis-tick arguments — it expects scalar range x=a..b. Replaced with matrixplot(M, ...) which renders a matrix as a colour-mapped surface and requires no explicit axis grids (rows/columns are the axes automatically).

Error 3 — zgradient color option inside surfdata.

color=zgradient([...]) is invalid syntax for surfdata/matrixplot. The correct Maple 2015+ syntax is: colorscheme=["zgradient", ["blue","cyan","green","yellow","red"]]

This fix is applied to both the f_1(r,t) and P(r,t) three-dimensional plots.
